ETS ridership surge predictor

Predicted departures for each district, with the reasons behind every prediction

Departure district Kampar

Date ‹ 06/05/2026 ›

Data available from Thursday, 1 January 2026 to Wednesday, 23 September 2026. None of it was used to train the model.

Departing from

Kampar Perak

Date

Wednesday, 6 May 2026

Predicted trips

224

Actual trips

173

Prediction over by 51 (29.7%)

Compared with a normal Wednesday

−2%

Normal demand

Whole ETS network that day: 9,354 trips predicted, 9,012 actual.

Why the model predicts this

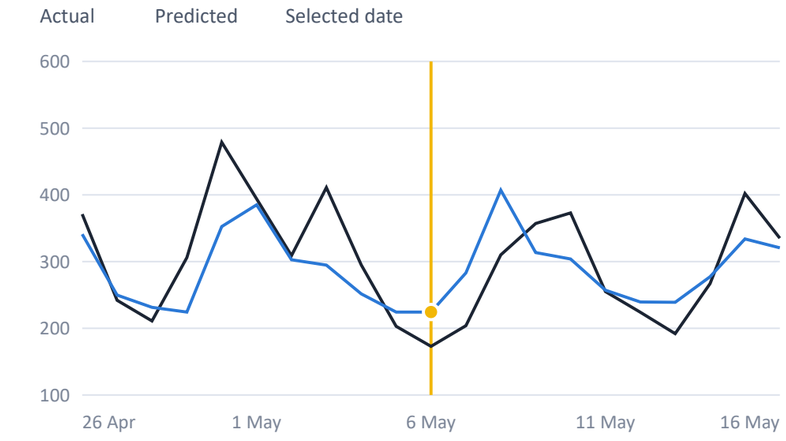
Main reasons for extra trips: Rainfall 28.0 mm; Average temperature 26.6 °C.

Normal level for Kampar (set by its population and income)		**282 trips**
Wednesday	−55	
Rainfall 28.0 mm	+8	
Average temperature 26.6 °C	+8	
21 days before a festival	−6	
Outside festive travel periods	−6	
No festival in the past 30 days	−3	
Other conditions (5)	−3	

282 normal level −57 from today's conditions = 224 trips predicted

Each bar is a TreeSHAP value: how many trips that condition adds to or removes from the district's normal level. Normal level + all bars = the prediction.

Three weeks around this date



Over these 21 days the prediction was off by 46 trips a day on average (15.2% of actual demand).

Biggest predicted surge in each district, 2026

Each district's busiest predicted day, ranked by extra trips above a normal day of the same weekday: the additional passengers the station must handle. Select one to inspect it.

Kuala Lumpur **+1,393 trips (+42%)**
Tuesday, 17 February 2026, Chinese New Year

Kinta **+426 trips (+27%)**
Wednesday, 18 February 2026, Chinese New Year

Perlis **+336 trips (+55%)**
Wednesday, 25 March 2026, Hari Raya Aidilfitri

Seberang Perai Utara **+275 trips (+38%)**
Wednesday, 27 May 2026, Hari Raya Haji

Petaling **+250 trips (+67%)**
Wednesday, 27 May 2026, Hari Raya Haji

Larut dan Matang **+222 trips (+65%)**
Wednesday, 18 February 2026, Chinese New Year

Kota Setar **+194 trips (+78%)**
Monday, 23 March 2026, Hari Raya Aidilfitri

Seberang Perai Tengah **+169 trips (+45%)**
Wednesday, 18 February 2026, Chinese New Year

Kuala Muda **+168 trips (+59%)**
Sunday, 22 March 2026, Hari Raya Aidilfitri

Tampin **+145 trips (+110%)**
Wednesday, 18 February 2026, Chinese New Year

Batang Padang **+138 trips (+82%)**
Thursday, 21 May 2026, Hari Raya Haji

Kerian **+137 trips (+89%)**
Wednesday, 18 February 2026, Chinese New Year

Model: Random Forest (100 trees) implemented in NumPy, trained on ETS ridership from 2023 to 2025 with weather, public holidays, festivals and DOSM district data. Explanations: TreeSHAP. On the unseen 2026 data the network daily R^2 is 0.304 and the festive-day error is 15.75%, against 21.75% for a simple same-weekday average. Data prepared 2026-10-05 07:25.